bi_scope_certificates: covering models that are nearly the same

What this folder shows

A BI value for a set of models covers only those models. The paper extends it to every model whose scores stay within a small distance of one model in the set. The idea is simple: if the winning score of a model leads the runner-up by more than that distance, any nearby model makes the same guess. So the certificate is again a count: the traces where the model's lead is large enough.

Script	What it produces	Used for
`compute_full_suite_bi_loc.py`	these counts for all 80 models of one trace set, at nine distances	Table 9
`profile_epsilon_calibration.py`	the counts on two parts of the traces, plus how much the scores move when the model is run again, in another batch size, in lower precision, or with noise	Figure 2
`compute_bi_loc_margin_curves.py`	the same curves for chosen models on all their attack traces	numbers in Section 5.3
`prepare_aesrd_estranet_h5.py`	the input file for the AES-RD Transformer	Level C
`build_bi_scope_tables.py`	tables from the curves	long version

Before you start

The Docker image (or the Python environment with `requirements-ml.txt`), the TCHES20 download, and the file `results/paper_results/bi_suite_dataset_summary.csv`, which Level A or B writes. The Transformer models also need the source code and models from guide 01, Level C.

Run

Counts for the 80 ASCAD (desync 50) models at nine distances:

```
python -m bi_scope_certificates.compute_full_suite_bi_loc --dataset ascad_desync_50 \
  --suite-csv results/paper_results/bi_suite_dataset_summary.csv \
  --output-dir results/bi_scope_certificates/full_suite_loc
```

```
$ python -m bi_scope_certificates.compute_full_suite_bi_loc --dataset ascad_desync_50 --suite-csv results/pa
    per_results/bi_suite_dataset_summary.csv --output-dir results/bi_scope_certificates/full_suite_loc
[... earlier output omitted; key lines kept ...]
[...]
[52/80] loading zaid_ascad_desync_50_horizontal_scaling_m1_1_1
[53/80] loading zaid_ascad_desync_50_horizontal_scaling_m1_1_2
[54/80] loading zaid_ascad_desync_50_horizontal_scaling_m1_1_3
[55/80] loading zaid_ascad_desync_50_horizontal_scaling_m1_1_4
[56/80] loading zaid_ascad_desync_50_horizontal_scaling_m1_1_5
[57/80] loading zaid_ascad_desync_50_horizontal_scaling_m1_1_6
[58/80] loading zaid_ascad_desync_50_horizontal_scaling_m1_1_7
[59/80] loading zaid_ascad_desync_50_horizontal_scaling_m1_1_8
[60/80] loading zaid_ascad_desync_50_horizontal_scaling_m1_1_9
[61/80] loading zaid_ascad_desync_50_horizontal_standardization_0
```

```
[62/80] loading zaid_ascad_desync_50_horizontal_standardization_1
[63/80] loading zaid_ascad_desync_50_horizontal_standardization_2
[64/80] loading zaid_ascad_desync_50_horizontal_standardization_3
[65/80] loading zaid_ascad_desync_50_horizontal_standardization_4
[66/80] loading zaid_ascad_desync_50_horizontal_standardization_5
[67/80] loading zaid_ascad_desync_50_horizontal_standardization_6
[68/80] loading zaid_ascad_desync_50_horizontal_standardization_7
[69/80] loading zaid_ascad_desync_50_horizontal_standardization_8
[70/80] loading zaid_ascad_desync_50_horizontal_standardization_9
[71/80] loading zaid_ascad_desync_50_no_preprocessing_0
[72/80] loading zaid_ascad_desync_50_no_preprocessing_1
[73/80] loading zaid_ascad_desync_50_no_preprocessing_2
[74/80] loading zaid_ascad_desync_50_no_preprocessing_3
[75/80] loading zaid_ascad_desync_50_no_preprocessing_4
[76/80] loading zaid_ascad_desync_50_no_preprocessing_5
[77/80] loading zaid_ascad_desync_50_no_preprocessing_6
[78/80] loading zaid_ascad_desync_50_no_preprocessing_7
[79/80] loading zaid_ascad_desync_50_no_preprocessing_8
[80/80] loading zaid_ascad_desync_50_no_preprocessing_9
results/bi_scope_certificates/full_suite_loc/ascad_desync_50_full_suite_bi_loc_summary.csv
[exit code 0, 1 min 9 s]
```

```
$ cat results/bi_scope_certificates/full_suite_loc/ascad_desync_50_full_suite_bi_loc_summary.csv
dataset,scope,M,epsilon,n_te,S_loc_max,p_hat_loc_max,p_loc_plus,BI_obs_loc,BI_loc_plus,BI_suite_plus,active_
    center,active_architecture,score_normalization,A,delta_effective,active_margin_min,active_ma …
ascad_desync_50,full TCHES20-M80 score-local suite,80,0.0,10000,97,0.0097,0.017350003293937932,1.31220046263
    76784,2.151079746927801,2.122567249526002,zaid_ascad_desync_50_no_preprocessing_4,CNN,probab …
ascad_desync_50,full TCHES20-M80 score-local suite,80,1e-05,10000,98,0.0098,0.017481236708476178,1.326997464
    5657587,2.1619510620015916,2.122567249526002,zaid_ascad_desync_50_no_preprocessing_4,CNN,pro …
ascad_desync_50,full TCHES20-M80 score-local suite,80,3e-05,10000,100,0.01,0.017743219068263653,1.3561438102
    252754,2.183411584943912,2.122567249526002,zaid_ascad_desync_50_no_preprocessing_4,CNN,proba …
ascad_desync_50,full TCHES20-M80 score-local suite,80,0.0001,10000,112,0.0112,0.019302476835074082,1.5196425
    425081548,2.3049297918331413,2.122567249526002,zaid_ascad_desync_50_no_preprocessing_4,CNN,p …
ascad_desync_50,full TCHES20-M80 score-local suite,80,0.0003,10000,141,0.0141,0.022996124246323305,1.8518389
    72849344,2.5575345408827825,2.122567249526002,zaid_ascad_desync_50_no_preprocessing_4,CNN,pr …
ascad_desync_50,full TCHES20-M80 score-local suite,80,0.001,10000,815,0.0815,0.1001108889007633,4.3829438695
    6899,4.679670807445984,2.122567249526002,zaid_ascad_desync_50_no_preprocessing_2,CNN,probabi …
ascad_desync_50,full TCHES20-M80 score-local suite,80,0.003,10000,8305,0.8305,0.8535488309074055,7.732052073
    300289,7.77154559625994,2.122567249526002,zaid_ascad_desync_50_no_preprocessing_2,CNN,probab …
ascad_desync_50,full TCHES20-M80 score-local suite,80,0.01,10000,10000,1.0,1.0,8.0,8.0,2.122567249526002,zai
    d_ascad_desync_50_no_preprocessing_2,CNN,probability_simplex_A=1,1.0,1.3888888888888888e-09, …
ascad_desync_50,full TCHES20-M80 score-local suite,80,0.03,10000,10000,1.0,1.0,8.0,8.0,2.122567249526002,noC
    onv1_ascad_desync_50_no_preprocessing_8,MLP,probability_simplex_A=1,1.0,1.3888888888888888e- …
[exit code 0, 0 s]
```

The two public models of the same trace set, with the traces split 20% for choosing the distance and 80% for the certificate:

```
python -m bi_scope_certificates.profile_epsilon_calibration --dataset ascad_desync_50 --public-only \
  --output-dir results/bi_scope_certificates/calibration/ascad_desync_50
```

```
$ python -m bi_scope_certificates.profile_epsilon_calibration --dataset ascad_desync_50 --public-only --outp
    ut-dir results/bi_scope_certificates/calibration/ascad_desync_50
Creating all target labels for the attack traces:
[noConv1_ascad_desync_50_no_preprocessing_0] baseline n=10000 cal=2000 eval=8000
Creating all target labels for the attack traces:
[zaid_ascad_desync_50_no_preprocessing_3] baseline n=10000 cal=2000 eval=8000
{"centers": 2, "dataset": "ascad_desync_50", "errors": 0}
[exit code 0, 10 s]
```

```
$ ls results/bi_scope_certificates/calibration/ascad_desync_50 && \
    head -12 results/bi_scope_certificates/calibration/ascad_desync_50/epsilon_variant_distances.csv
epsilon_margin_counts.csv
epsilon_variant_distances.csv
manifest.json
profile_noConv1_ascad_desync_50_no_preprocessing_0.npz
profile_zaid_ascad_desync_50_no_preprocessing_3.npz
```

```
dataset,architecture,center_id,variant,variant_group,score_normalization,A,n_cal,distance_mean,distance_p99,
    distance_p999,distance_max,coverage_1e-5,coverage_3e-5,coverage_1e-4,coverage_3e-4,coverage_ …
ascad_desync_50,MLP,noConv1_ascad_desync_50_no_preprocessing_0,repeat_base_batch,fp32_repro,probability_simp
    lex_A=1,1.0,2000,0.0,0.0,0.0,0.0,0.0,1.0,1.0,1.0,1.0,1.0
ascad_desync_50,MLP,noConv1_ascad_desync_50_no_preprocessing_0,batch_64,fp32_repro,probability_simplex_A=1,1
    .0,2000,0.0,0.0,0.0,0.0,1.0,1.0,1.0,1.0,1.0
ascad_desync_50,MLP,noConv1_ascad_desync_50_no_preprocessing_0,batch_2048,fp32_repro,probability_simplex_A=1
    ,1.0,2000,0.0,0.0,0.0,0.0,1.0,1.0,1.0,1.0,1.0
ascad_desync_50,MLP,noConv1_ascad_desync_50_no_preprocessing_0,input_fp16_storage,fp16_storage,probability_s
    implex_A=1,1.0,2000,0.0,0.0,0.0,0.0,1.0,1.0,1.0,1.0,1.0
ascad_desync_50,MLP,noConv1_ascad_desync_50_no_preprocessing_0,score_fp16_storage,fp16_storage,probability_s
    implex_A=1,1.0,2000,4.7079073265194895e-06,1.4130957424640656e-05,1.6112249344587316e-05,2.7 …
ascad_desync_50,MLP,noConv1_ascad_desync_50_no_preprocessing_0,score_quant_12bit,score_quantization,probabil
    ity_simplex_A=1,1.0,2000,0.0001286718572872781,0.0001788132175031428,0.0002540557722932317,0 …
ascad_desync_50,MLP,noConv1_ascad_desync_50_no_preprocessing_0,score_quant_13bit,score_quantization,probabil
    ity_simplex_A=1,1.0,2000,6.423572543374916e-05,8.602533665567051e-05,0.0001201596001167901,0 …
ascad_desync_50,MLP,noConv1_ascad_desync_50_no_preprocessing_0,score_quant_14bit,score_quantization,probabil
    ity_simplex_A=1,1.0,2000,3.201166275038579e-05,4.0278899763194137e-05,5.603660862280591e-05, …
ascad_desync_50,MLP,noConv1_ascad_desync_50_no_preprocessing_0,score_quant_16bit,score_quantization,probabil
    ity_simplex_A=1,1.0,2000,7.99038100383186e-06,1.008784107881893e-05,1.3958238756522137e-05,1 …
ascad_desync_50,MLP,noConv1_ascad_desync_50_no_preprocessing_0,input_quant_8bit,input_quantization,probabili
    ty_simplex_A=1,1.0,2000,0.000977531326469034,0.004363322183489798,0.007841318430378871,0.014 …
ascad_desync_50,MLP,noConv1_ascad_desync_50_no_preprocessing_0,input_quant_10bit,input_quantization,probabil
    ity_simplex_A=1,1.0,2000,0.00023766890750266613,0.0011626615747809407,0.001969645865261544,0 …
[exit code 0, 0 s]
```

What to look for

- The summary has one row per distance: the largest count over the 80 models, the matching rate, the limit in bits, and which model gave it. At distance 0 the count is the plain number of correct guesses. At the largest distance it covers almost every trace, which is why the table in the paper stops there.
- `epsilon_margin_counts.csv` holds the counts for both parts of the traces. The Figure 2 script picks the smallest distance that changes the count of at least 75% of the twelve models and is larger than the score movement seen when the models are rerun.
- `epsilon_variant_distances.csv` shows that movement. On a CPU, running a model again gives the same scores to within 0.000001. On the GPU used for the paper, the Transformer scores moved by up to 0.00019, which is why the paper's distance is 0.0003.